

A new static shift correction method of magnetotelluric: Based on the analysis of 3D forwarding results

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Summary

To solve the problem of magnetotelluric (MT) static shift correction, we quantize the influencing factors of geological environment based on three-dimensional (3-D) MT theory, and study the MT static shift by three-dimensional numerical simulation techniques. We find that the static shift distortions not only affect the apparent resistivity, but also affect the impedance phase. The results of distortion are related to the frequency as well. Based on the analysis on synthetic data, we propose the concept of the generalized static shift resistivity and a new correction method of MT static shift. The correct approach is verified by the study of two-dimensional (2-D) synthetic models.

Introduction

Magnetotelluric is one of the most important methods of deep crust exploration and prospecting. The static shift effect is a very common phenomenon in MT field data, which seriously impacts on the accuracy of data, and hence the interpretation of this method. Static shift is often considered to be generated from the secondary electric field caused from the accumulated charge on the boundary of two-dimensional or three-dimensional heterogeneous (C. Torres-Verdin and F. X. Bostick Jr., 1992; Park et.al, 1983, 1985; Wannamaker et.al, 1984). While the apparent resistivity curve is often moved by static shift effect on the frequency axis (Sternberg et.al, 1988), the impedance phase is mostly not changed (Jones, 1983).

There are five static shift correction approaches, namely the spatial filtering methods (Berdichevsky et.al, 1980; Sternberg et.al, 1982; Warner et.al, 1983; Bostick, 1986; Torres Verdin and Bostick, 1992; Liu, 2011), impedance tensor decomposition methods (Groom et.al, 1989; Yang, 2002), comprehensive data interpretation methods (Berdichevsky and Dmitriev, 1976; Larsen, 1977; Wannamaker et.al, 1984; Jones, 1988; deGroot-Hedli, 1991; Smith and Booker, 1991; Fang, 1993; Meju, 1996; Wang, 1998), independent measurement correction methods (Andrienx and Wightman, 1984; Sternberg et.al, 1988; Pellerin and Hohmann, 1990; Zhang et.al, 1995; Yao et.al, 2001; Tripaldi, 2010) and phase correction methods (Berdichevsky and Dmitriev, 1976; He, 1993; B. Tournier et.al, 2007).

The above concept and correction methods of static shift are all based on 1-D or 2-D MT theories and assumptions. However, that does not meet the definition of 3-D static shift effect. Therefore the conventional methods have the uncertainty for correction results (Liu, 2011). With 3-D theory considered, we divide the factors affecting static shift effect into two categories, geology factors and observation factors. For geology factors, we obtain the MT response of different models containing shallow heterogeneous bodies through 3-D MT forwarding program, determine the relationship of factors and static shift in quantum, and then try to find a new static shift correction method based on these correspondences.

The analysis of geology factors of static shift effect

The factors of the static shift effect can be divided into two categories, geology factors and observation factors. Geology factors consist of two parts: (1) the resistivity difference between the heterogeneous bodies and background, and the scale and depth of the heterogeneous bodies; (2) peripheral faults on surface.

Based on 3-D MT theory, we quantize the geology factors and study the relationships between these factors and MT static shift effect utilizing 3-D numerical simulation techniques (code by ZHANG Kun, 2012). Then we try to find a

new correction method of static shift by these relationships.

Here we take the resistivity factor as an example to explain the influence of geology factors on static shift. As shown in Figure 1 and 2, two surface heterogeneous bodies (both are of 4m*4m*4m scale) are set in a half-space model of 100 ohm m. The resistivity of the heterogeneous bodies varies from 1 to 10000. Through the 3-D forward scheme, MT sounding responses are generated. The relation curves of static shift against heterogeneous resistivity factor of apparent resistivity (figure 3) and impedance phase (figure 4) are hence obtained at a frequency range of 320Hz to 1Hz.

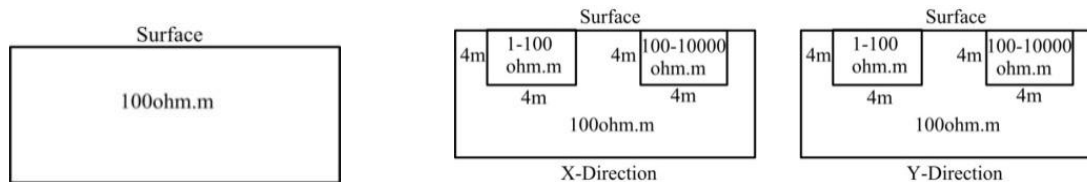


Fig.1 Half-space model

Fig.2 Resistivity changed model

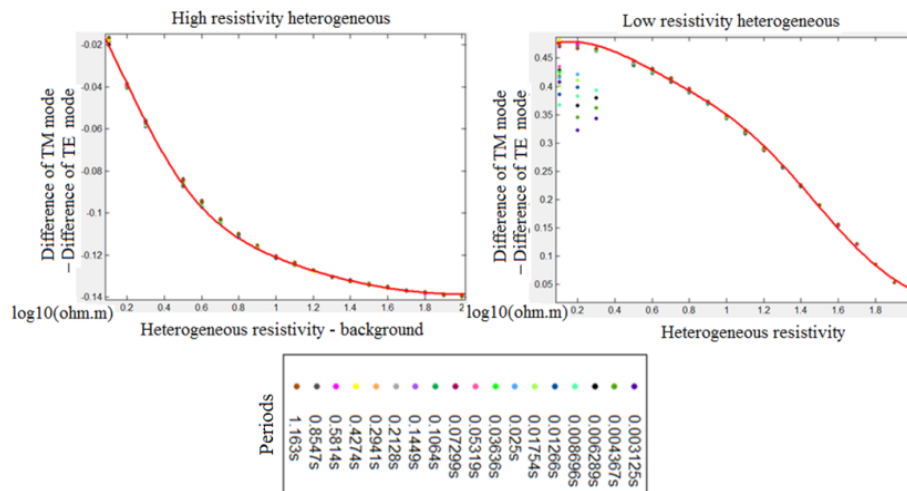


Fig.3 Apparent resistivity difference of TM mode and TE mode

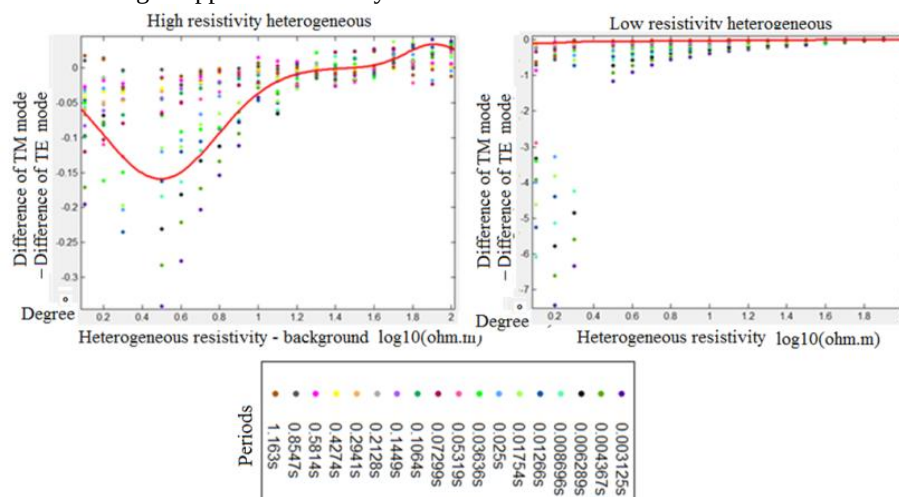


Fig.4 Impedance phase difference of TM mode and TE mode

First we define the “difference of TM mode” as the original apparent resistivity value of TM mode subtracts static shifted value of TM mode. We also define the “difference of TE mode” as regular value of TE mode subtracts static shifted value of TE mode. Then we define the concept of “quantity difference” as the difference of the formal two values. From figure 3 and 4, we can see that the apparent resistivity and phase quantity difference is changed as the resistivity of the heterogeneous bodies changes, of which the relationship submits to non-linear regulation. While the quantity difference varies little as the frequency changes, quantity difference of impedance phase changes a lot. This shows that phase is a better indicator for static shift than apparent resistivity.

Correction method of static shift

Geology factors can be generalized as the resistivity parameters and geometry parameters. We find that adding small electrical anomalies into the initial inversion model can solve the static shift problem. And the resistivity of the bodies can be given at discretion. The only thing to be determined is that the value should be smaller than background or not, which is defined as the generalized static shift resistivity. At the same time, geometry parameters of the anomalies need not to be considered.

(1) The determination method of generalized static shift resistivity

The generalized static shift resistivity must be determined according to the above-mentioned 3-D forwarding results of the geology factors. The specific procedures are as follows:

If the frequency-apparent resistivity curves of TE and TM mode are parted on high frequency band, the generalized static shift resistivity should be higher than the background resistivity when the quantity difference is less than zero. And the generalized static shift resistivity should be lower than the background resistivity when the quantity difference is greater than zero.

If the frequency-apparent resistivity curves of TE and TM mode coincides on high frequency band, put the impedance phase of two modes into first quadrant. The generalized static shift resistivity should be higher than background when the quantity difference is less than zero. And the generalized static shift resistivity should be lower than background when the quantity difference is greater than zero.

If the quantity difference of apparent resistivity and impedance phase is small, put the impedance phase of two modes into first quadrant. The generalized static shift resistivity should be higher than background resistivity when the quantity difference of phase approaches to zero from negative value as the frequency decreases. And the generalized static shift resistivity should be lower than background resistivity when the quantity difference of phase approaches to zero from positive value as frequency decreases. Quantity difference at a few frequencies may jump a lot. In that case, we need to use more frequencies, and judge prudently.

(2) 2-D inversion example

A series of models are designed to test the efficiency of the static correction method, as shown in figure 5. The upper-left plot shows a normal model (I) without any local heterogeneous. The resistivity of the two object bodies are 10 ohm m and 1000 ohm m respectively. The lower-left model is a static shifted model (II) with two local heterogeneous bodies (10ohm.m and 1000ohm.m). The upper right plot (A) is the inversion result of the synthetic data generated from model I. The middle-right plot (B) is the inversion result of the static shifted data generated from model II. The lower right plot (C) is the inversion result of the static shifted data from model II, but with an initial model modified by two local heterogeneous bodies with generalized static shift resistivity. The quantity differences of apparent resistivity is 1.5 and -0.4 for the two abnormal data site at 800m and 2300m in logarithmic domain, so the generalized static shift resistivity is set to be 1 ohm m and 10000 ohm m. The inversion is calculated by the 2-D NLCC inversion method proposed by Rodi and Mackie in 2001.

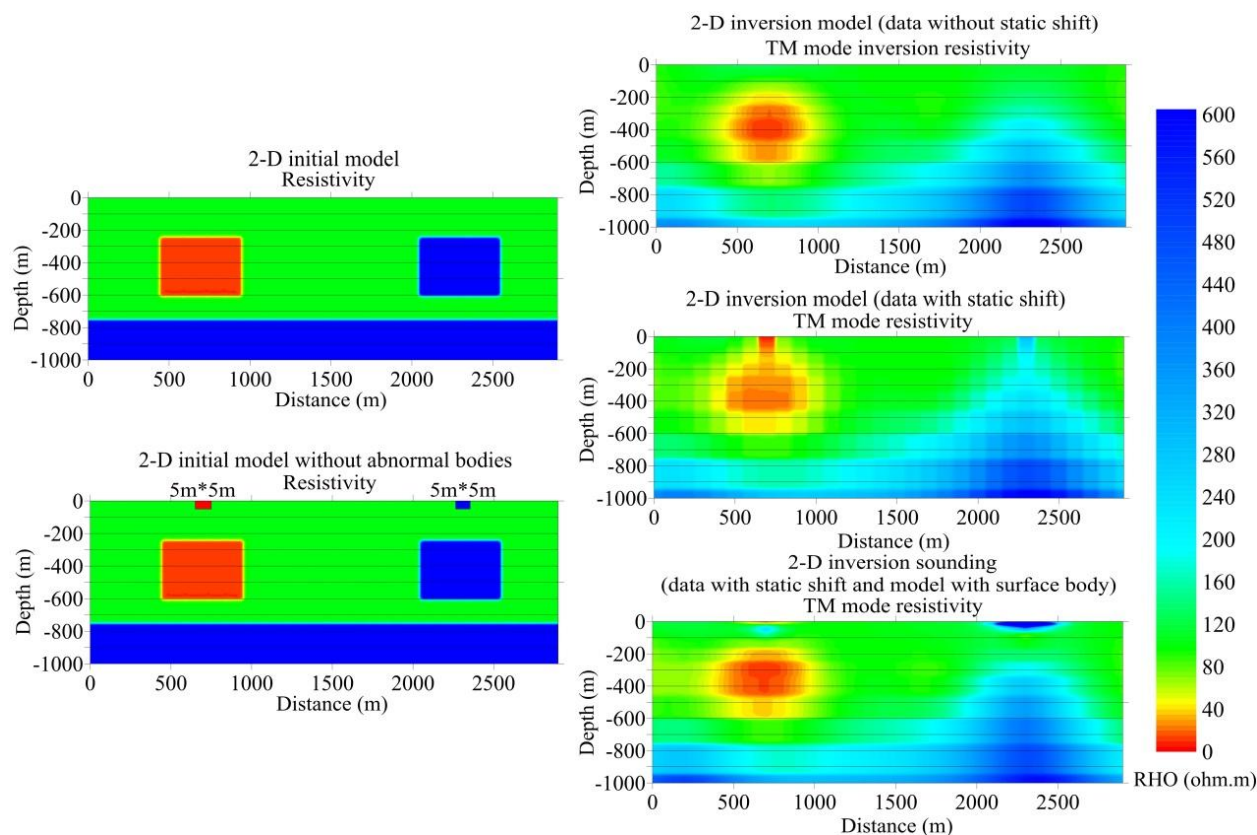


Fig.5 Original and inversion models

As shown in figure 5, the low and high resistivity “channel” observed in model B cannot be found in model C. The resistivity value and the scale of target bodies in model C are consistent with the “unshifted” model of A. So the correction method is effective. But the small high resistivity region below the heterogeneous body in model C should be ignored in geophysics and geology interpretation.

Conclusions

The relationship between the geological factors and the MT static shift effect is studied with 3-D numerical simulation techniques. A concept of generalized static shift resistivity is proposed to measure the influence of the static shift effect. And a new correction method of static shift is presented utilizing these relationships.